

Circulating Metabolic Markers Identify Patients at Risk for Tumor Recurrence

A Prospective Cohort Study in Colorectal Cancer Surgery

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Objective: To investigate the spermidine pathway capability to predict patients at risk for tumor recurrence following colorectal cancer (CRC) surgery.

Background: Recurrence rates after CRC surgery remain at about 20% despite an optimal technique and adjuvant therapy when necessary. Identification of risk biomarkers of recurrence is an unmet need. The spermidine pathway is indispensable for cell proliferation and differentiation, and is suggested to accelerate tumor spread.

Methods: This was a prospective cohort study of patients undergoing CRC surgery from 2015 to 2018. Plasma samples were collected before surgery and on postoperative day 4, and the spermidine pathway was assessed through mass spectrometry. Oncological outcomes were registered.

Results: A total of 146 patients were included and 24 (16.4%) developed tumor recurrence. Higher levels of preoperative spermidine pathway components (spermidine, spermine, spermidine synthase enzyme, and spermine/arginine balance) were positively

associated with recurrence. Surgery promoted a decrease in these pathway elements. The greater the decline was, the lower the risk of recurrence. Preoperative spermidine over the cut-off of 0.198 μ M displayed a 4.69-fold higher risk of recurrence. The spermine synthase enzyme behaved in the opposite direction.

Conclusions: The spermidine pathway is associated with tumor recurrence following CRC surgery and, after confirmation in larger cohorts, could be translated as a risk biomarker of recurrence into clinical practice.

Keywords: metabolomics, spermidine pathway, colorectal cancer, tumor recurrence

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Surgery is the treatment of choice for more than 80% of patients with colorectal cancer (CRC), with lymph node involvement being the most important prognostic factor.^{1,2} However, the 5-year risk of recurrence remains about 20%, despite a complete resection with an optimal lymphadenectomy and adjuvant treatment in stage III and high-risk stage II patients.^{3,4} Other tumor-related factors, such as lymphovascular or perineural invasion and degree of differentiation, have not improved our capacity to predict the individual oncological outcome.⁵ The study of other factors that may affect long-term cancer-specific results could provide information on the mechanisms of relapse, as well as biomarkers to identify patients at higher risk who could benefit from targeted therapies.

Our group has been investigating the inflammatory response to CRC and surgical trauma as a potential mechanism of tumor recurrence. Soluble factors released by the inflammatory response could potentially stimulate the survival, proliferation, and invasiveness of tumor cells through a combination of mechanisms, such as increased angiogenesis, impaired immune function, and induction of an epithelial-to-mesenchymal transition trait.^{6–9} Apart from an increased inflammatory response, other factors related to tumor progression could be involved in the recurrence. Tumor cells have been shown to display increased metabolic autonomy, taking up nutrients and metabolizing them in pathways that support growth and proliferation; increasing not only bioenergetic but also biosynthetic activities. The activation and formation of metabolic networks with stromal cells produce energy-rich metabolites that fuel tumor growth.^{10–12}

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In CRC, the spermidine pathway has long been associated with tumor development and growth.¹³ Biosynthesis of spermidine begins with arginase catalyzing the conversion from arginine to ornithine, then the decarboxylation of ornithine to putrescine by ornithine decarboxylase. Finally, the polyamines spermidine and spermine are produced. Spermidine is created by the action of spermidine synthase and spermine is formed from spermidine under the catalysis of spermine synthase (see Figure, Supplemental Digital Content 1, <http://links.lww.com/SLA/F218>, which demonstrates the biosynthesis of spermidine).¹⁴ Interestingly, polyamine production is greater in rapidly growing cells as they are indispensable for cell proliferation and differentiation, being one of the first events in proliferating cells that precede both nucleic acid and protein syntheses. They have also been proposed to accelerate tumor spread by decreasing cancer cell adhesion, stimulating angiogenesis, and favoring immune evasion.¹⁵

In this sense, the spermidine pathway has already been suggested to support traditional diagnostic tools for the accurate diagnosis of CRC. Plasma levels of polyamines and related amino acids could be useful in discriminating CRC from healthy patients.^{16,17} Moreover, Weiss et al¹⁸ found that T3 and T4 CRC tumors have greater total polyamine and spermine concentrations than T2 tumors, implying that higher polyamines may be associated with a higher tumor stage. Another group introduced that changes in polyamine metabolism could be related not only to the CRC stage but also to its infiltration level and prognosis,¹⁹ opening up the possibility that the spermidine pathway could be involved in the oncological outcome as well. Nevertheless, the role of the spermidine pathway in CRC recurrence has hardly been studied.

In the present study, our aim was to investigate the spermidine pathway capability to predict patients at risk for tumor recurrence after CRC surgery. We evaluated the associations of preoperative and postoperative components related to this pathway, with relapse. We hypothesized that the spermidine pathway would reliably predict the risk of recurrence, which would be useful in clinical practice.

METHODS

Study Design and Participants

This observational single-center study included a prospective cohort of patients who underwent CRC surgery between October 2015 and May 2018. The study was approved by the Ethical Committee and was registered at ClinicalTrials.gov (NCT02789709). All patients provided written informed consent. Patients with stage IV cancer, receiving neoadjuvant therapy, noncurative resection, or urgent surgery were excluded.

Patient characteristics including age, sex, and body mass index were collected. For predicting operative risk, the American Society of Anesthesiologists (ASA) physical status score was used. The preoperative relevant blood test result was carcinoembryonic antigen (CEA). Perioperative outcomes were registered, along with surgical approach, tumor location, and intra-abdominal infection. Pathologic data consisted of tumor stage according to TNM, histologic type, and differentiation. Adjuvant chemotherapy treatment and regimen were also gathered. Oncological outcomes data comprised recurrence and its type (local and/or systemic).

Patients were followed every 3 months during the first 2 years and then every 6 months until 5 years. At each

follow-up visit, a full history, physical examination, and blood test including CEA were performed. A computed tomography scan was accomplished every 6 months in stage III patients and those with relapse suspicion. Colonoscopy was performed 1 year after surgery and then every 3 years.

Spermidine Pathway

Venous blood samples were collected before surgery (baseline) and on postoperative day 4, following a 6-hour period of fasting and first thing in the morning, to minimize interpatient metabolic variations related to dietary intake and circadian rhythm. The plasma spermidine pathway was assessed through the quantification of arginine, ornithine, putrescine, spermidine, and spermine using the AbsoluteIDQ p180 kit (Biocrates Life Sciences AG, Innsbruck, Austria). Enzymatic activity of arginase, ornithine decarboxylase, spermidine synthase, and spermine synthase, along with spermine/arginine balance, were also estimated. The sample plate was processed according to the manufacturer's instructions. Analyses were performed in the AB Sciex 4000 QTRAP MS/MS mass spectrometer (AB Sciex LLC, Framingham, MA) coupled to an Agilent 1260 Infinity HPLC system (Agilent, Santa Clara, CA). Analyst (Sciex) and the MetIDQ software packages were used to analyze the data and calculate metabolite concentrations, as stated in μM .

Statistical Analysis

Data preprocessing and analysis were conducted using the R software version 4.2.²⁰ Demographic and clinical differences between groups were assessed with data collected at the onset of the study. Pearson χ^2 , independent Student test, or Wilcoxon tests were used as appropriate. Values of metabolic markers were log-transformed to enhance the normality of distributions. Longitudinal associations between the spermidine pathway components before surgery as well as the effect of surgery (defined as Δ -day 4 minus baseline data), and the risk of recurrence, were analyzed through weighted Cox regression using the R package "coxphw".²¹ Multivariate weighted Cox models were adjusted for age, sex, TNM stage, histologic differentiation, intra-abdominal infection, and preoperative CEA concentrations. Time was set as the number of days between sampling and relapse. Corrected *P* values were calculated through the Benjamini-Hochberg procedure²² to control for the false discovery rate (FDR). Polar plots were acquired illustrating the hazard ratios (HR) and the 95% confidence intervals (CIs) of each metabolic marker [HR (95% CI)]. The receiver operating characteristic (ROC) analyses of the baseline significant markers were performed, and the optimal concentrations that maximize the sensitivity and specificity of ROC curves were calculated using the R package "OptimalCutpoints".²³ To investigate the translational application of such metabolic markers, optimal concentration cut-offs were employed to establish a risk threshold. After dichotomizing markers based on their cut-off values, the risk over time was evaluated through Kaplan-Meier survival curve estimations with the regular log-rank test method for comparison. A *P* value <0.05 was considered significant.

RESULTS

Study Participants

During the study interval, 150 individuals (59 females and 91 males) were diagnosed with nonmetastatic CRC and considered eligible for the study. Four patients were excluded due to loss of follow-up. Patient, surgical, and pathologic characteristics are shown in Table 1. Adjuvant chemotherapy was administered to 42 (28.8%) patients. Treatment regimens included FOLFOX (folinic acid, 5-fluorouracil, and oxaliplatin), CAPOX (capecitabine and oxaliplatin), and capecitabine monotherapy, among others. After a minimum follow-up of 5 years, 24 (16.4%) patients developed tumor recurrence; local recurrence in 2 patients, systemic in 19, and local with systemic in 3.

Perioperative Spermidine Pathway is Associated With CRC Recurrence

Results from multivariate weighted Cox models denoted the association of baseline spermidine pathway components and CRC recurrence, as shown in Table 2 and Figure 1A. Higher levels of spermidine (adjusted HR = 2.85, 95% CI = 1.50–5.41; *P* = 0.001), spermine (adjusted HR = 1.69, 95% CI = 1.16–2.45; *P* = 0.006), spermine/arginine balance (adjusted HR = 1.40, 95% CI = 1.08–1.80; *P* = 0.010), and spermidine synthase enzyme (adjusted

HR = 1.45, 95% CI = 1.02–2.06; *P* = 0.036) were positively associated with CRC relapse. On the contrary, a lower activity of spermine synthase was associated with recurrence (adjusted HR = 0.16, 95% CI = 0.04–0.62; *P* = 0.008).

Surgery promoted a decrease in the spermidine pathway components on postoperative day 4. Multivariate weighted Cox models on Δ-data (day 4 minus baseline) exhibited that, relative to the baseline, the greater the Δ-spermidine pathway elements following surgery, the lower the risk of recurrence, as shown in Table 2 and Figure 1B. Changes in spermidine (adjusted HR = 0.59, 95% CI = 0.44–0.80; *P* < 0.001), spermine (adjusted HR = 0.69, 95% CI = 0.55–0.88; *P* = 0.002), spermine/arginine balance (adjusted HR = 0.74, 95% CI = 0.59–0.93; *P* = 0.009), and spermidine synthase enzyme (adjusted HR = 0.75, 95% CI = 0.59–0.95; *P* = 0.015) were associated with CRC relapse. Likewise, the spermine synthase was the only metabolic marker that behaved in the opposite direction from the others (adjusted HR = 3.65, 95% CI = 1.26–10.60; *P* = 0.017).

Predicting CRC Recurrence Based on Preoperative Concentrations of Spermidine Pathway Components

The threshold concentrations of significant metabolic markers that simultaneously maximized both sensitivity and specificity were determined, as shown in Table 3. The Kaplan-Meier survival curve estimations showed that patients with preoperative levels above or below the threshold concentrations displayed an increased CRC recurrence, as shown in Table 3 and Figure 2 (see also Table, Supplemental Digital Content 2, <http://links.lww.com/SLA/F219>, which demonstrates patients' stratification based on these cut-off concentrations). Patients with preoperative spermidine concentration above 0.198 μM resulted in a 4.69-fold higher rate of relapse (unadjusted HR = 4.69, 95% CI = 1.95–11.32; *P* < 0.001). A spermine concentration above 0.201 μM exhibited an acceptable statistical performance with a positive association with recurrence (unadjusted HR = 2.80, 95% CI = 1.20–6.55; *P* = 0.013). However, the stratification of other markers did not achieve significance utilizing the described methodology.

DISCUSSION

The present study provides new insights into the relationship between the spermidine pathway and tumor recurrence in CRC patients. We have identified associations between the components related to this pathway and relapse following CRC surgery. Therefore, the results are consistent with the hypothesis that the spermidine pathway is not only associated with CRC diagnosis,^{16,17} but also with its recurrence. These findings have the potential to contribute to the identification of patients at high risk for relapse who would benefit from strict follow-up strategies, neo/adjuvant, and/or targeted therapies.

Derangements from normal metabolic pathways, which result in distinct products of altered metabolism, are widely established to be the hallmarks of a disease state. As a result, one potential clinical application of metabolomics in cancer is for the oncological outcome, providing a possible important tool for risk stratification and disease monitoring after treatment.²⁴ Specifically, the spermidine pathway has been proposed to play a role in several of those

TABLE 1. Patient, Surgical, and Pathologic Characteristics	
	N = 146
Age, y	71.2 ± 12.2
Sex, female/male	57 (39.0)/89 (61.0)
Body mass index, kg/m ²	27.4 ± 4.3
ASA physical status score	
I	3 (2.1)
II	71 (48.6)
III	71 (48.6)
IV	1 (0.7)
Preoperative CEA concentration, ng/mL	5.5 ± 9.4
Surgical approach	
Laparoscopy	120 (82.2)
Laparotomy	15 (10.3)
Laparoscopy converted to laparotomy	11 (7.5)
Tumor location	
Right and transverse colon	59 (40.4)
Left colon	55 (37.7)
Rectum	24 (16.4)
Synchronous	8 (5.5)
Intra-abdominal infection	
Abscess	16 (11.0)
Anastomotic leak	8 (5.5)
TNM stage	
Stage I	39 (26.7)
Stage II	62 (42.5)
Stage III	44 (30.1)
Stage IV (with curative surgery)	1 (0.7)
Histologic type	
Intestinal adenocarcinoma	135 (92.5)
Mucinous adenocarcinoma	8 (5.5)
Other	3 (2.1)
Histologic differentiation	
Grade 1–2 (G1–2)	129 (88.4)
Grade 3 (G3)	11 (7.5)
Not applicable	6 (4.1)

Values are either mean ± SD or N (%).
ASA indicates American Society of Anesthesiologists; CEA, carcinoembryonic antigen.

TABLE 2. Spermidine Pathway Associations With Colorectal Cancer Recurrence Using Baseline and Δ-data

Spermidine pathway	Baseline			Δ-data		
	HR (95% CI)	P	FDR	HR (95% CI)	P	FDR
Spermidine	2.850 (1.502-5.410)	0.001	0.012	0.593 (0.438–0.802)	< 0.001	0.006
Spermine	1.689 (1.163-2.454)	0.006	0.023	0.694 (0.549–0.878)	0.002	0.011
Spermine synthase	0.158 (0.040-0.624)	0.008	0.023	3.651 (1.257–10.603)	0.017	0.031
Spermine/arginine	1.397 (1.082-1.803)	0.010	0.023	0.739 (0.589–0.929)	0.009	0.028
Spermidine synthase	1.453 (1.024-2.060)	0.036	0.065	0.749 (0.592–0.946)	0.015	0.031
Arginine	0.592 (0.295-1.189)	0.140	0.211	1.730 (0.867–3.452)	0.120	0.180
Ornithine	1.931 (0.679-5.492)	0.217	0.279	0.529 (0.201–1.396)	0.198	0.255
Putrescine	1.230 (0.569-2.659)	0.599	0.674	1.072 (0.749–1.535)	0.703	0.703
ODT	0.981 (0.540-1.785)	0.951	0.951	1.159 (0.850–1.580)	0.351	0.395

HRs (95% CI) and *P* values from multivariate weighted Cox models adjusted for age, sex, TNM stage, histologic differentiation, intra-abdominal infection, and preoperative carcinoembryonic antigen (CEA) concentrations. Δ-data was calculated by subtracting the baseline from the postoperative day 4 data. ODT indicates ornithine decarboxylase.

known as hallmarks of cancer, including cell proliferation, differentiation, and tumor spread process by decreasing cancer cell adhesion, stimulating angiogenesis, and favoring immune evasion.^{15,25} We found that higher levels of spermidine, spermine, spermine/arginine balance, and spermidine synthase before surgery, and their decrease after surgery, were associated with CRC relapse; observed in the 16.4% of patients and consistent with the most recent published results, which demonstrate a decrease in the percentage of recurrence after CRC surgery with curative intent.⁴ Our results agree with those of Linsalata et al,²⁶ who divided 111 CRC patients into 4 quartiles according to their spermine levels inof tumor tissue and discovered that the HR of recurrence was 3.01 and 3.44 times higher in the second and third quartiles than in the lowest quartile, respectively. Our work extends their scope by providing the cut-off concentrations of the metabolic markers, which optimally differed according to the risk of presenting recurrence. This could offer a straightforward way to identify the high-risk patients, without dividing them into quartiles.

In the present study, surgery substantially decreased the levels of the spermidine pathway components. These decreases were observed on postoperative day 4. Other authors have also proposed a reduction in the concentrations of these pathway elements after CRC surgery, especially in putrescine.²⁷ However, the levels were reduced on the first day after surgery as compared with preoperative serum samples, implying that changes on the spermidine pathway are mainly influenced by surgical performance and thus suggesting that the effect of surgery occurs very early. Furthermore, we discovered that the greater the decline in the spermidine pathway components after CRC surgery compared with before surgery, the lower the risk of recurrence. This would be particularly interesting for disease monitoring. The investigation of the spermidine pathway levels at different time points during the follow-up will determine its usefulness for early detection of CRC relapse, as is the case with other biomarkers such as the CEA.²⁸ On the other hand, reduced spermine synthase activity was related to CRC recurrence in our work. Spermine synthase is the enzyme involved in the synthesis of spermine from

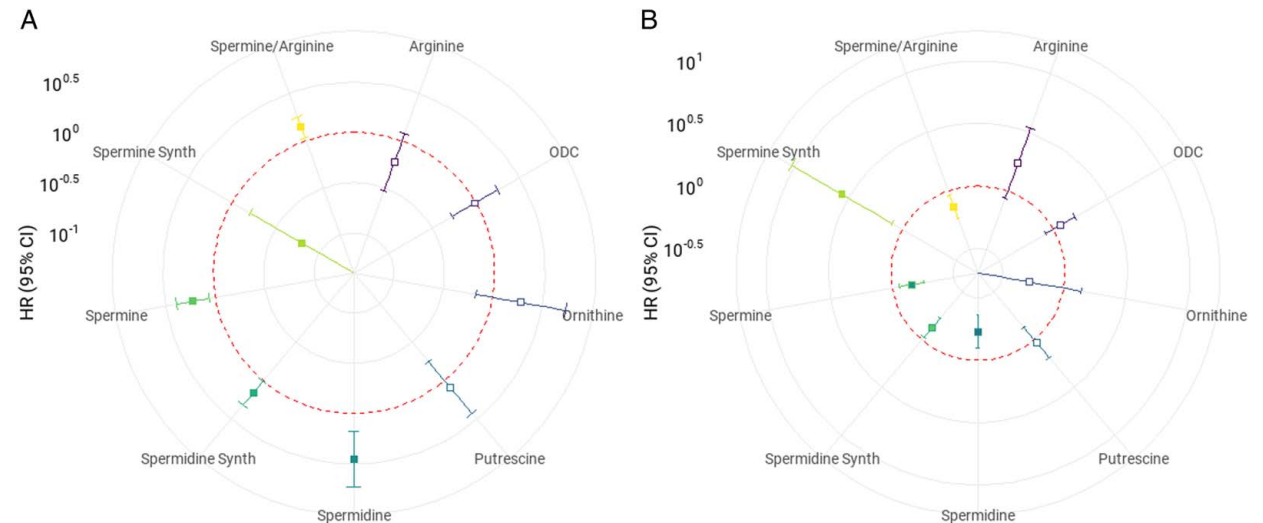


FIGURE 1. Polar plots illustrating HRs and 95% CI of spermidine pathway in baseline (A) and Δ-data (B), from multivariate weighted Cox models adjusted for age, sex, TNM stage, histological differentiation, intraabdominal infection, and preoperative carcinoembryonic antigen (CEA) concentrations. The left legend displays the HR values (scaled to log10) in the corresponding inner circle. The dotted circle indicates an HR of 1. ODC indicates ornithine decarboxylase; Synth, synthase.

TABLE 3. Likelihood of Colorectal Cancer Recurrence Relative To Baseline Spermidine Pathway Dichotomized By Cut-off Concentrations

Spermidine pathway	Cut-off	Units	HR (95% CI)	P
Spermidine	≥ 0.198	μM	4.69 (1.945–11.32)	< 0.001
Spermine	≥ 0.201	μM	2.80 (1.20–6.55)	0.013
Spermine synthase	≥ 0.995	μM/μM	0.48 (0.214–1.08)	0.071
Spermine/arginine	≥ 0.004	μM/μM	1.92 (0.85–4.32)	0.110
Spermidine synthase	≥ 2.497	μM/μM	1.77 (0.788–3.99)	0.160

HRs (95% CI) and P values from Kaplan-Meier survival curve estimations with the regular log-rank test method.

spermidine,¹⁴ and spermidine showed the strongest positive relationship with relapse. This could reflect an accumulation of spermidine compared with spermine in high-risk patients.

As mentioned above, the associations between the spermidine pathway components and CRC recurrence were already present in preoperative samples. Other groups detected characteristic metabolic signatures in CRC tissue samples that differentiated tumors according to T-stage²⁹ and N-stage.³⁰ Therefore, the prognostic value of this deregulation of metabolic markers before surgery points to an additional tool for an accurate preoperative assessment of high-risk patients. It has recently been demonstrated that 6 weeks of neoadjuvant chemotherapy for operable colon cancer may result in histopathologic downstaging, fewer incomplete resections, and improved 2-year disease control.³¹ Nevertheless, preoperative staging remains largely reliant on radiologic assessment, which could expose low-risk patients to the adverse events of chemotherapy.²⁹ On the basis of the present results, we propose that metabolic phenotyping, and the spermidine pathway in particular, could be a powerful method to reliably obtain a complex molecular representation of the tumor phenotype and guide

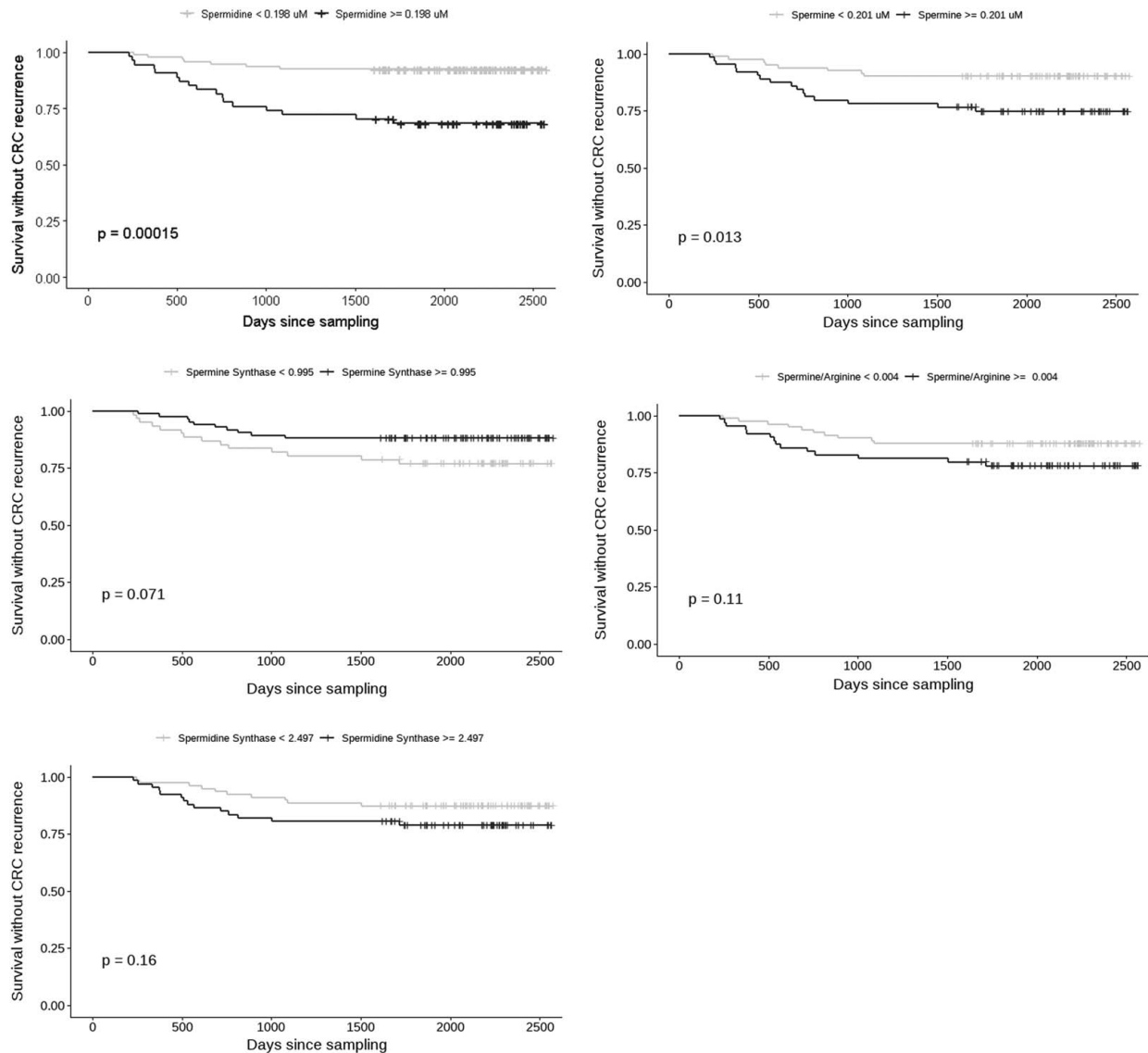


FIGURE 2. Kaplan-Meier survival curves for colorectal cancer recurrence along time according to optimized threshold concentrations of baseline spermidine pathway.

the appropriate treatment planning, including the identification of patients who would benefit from neoadjuvant treatment.

The detection of a concentration of the spermidine pathway components higher than the established cut-offs in a single sample should facilitate the application of our findings into clinical practice and may be the most suitable approach for confirmation in larger independent cohorts. Nonetheless, many clinical laboratories may not be able to benefit from these findings without access to this technology. Although mass spectrometry is a well-established technique that can provide cutting-edge solutions to clinical diagnostics and laboratory medicine to improve patient welfare,³² transforming the present results into a quick, simple, and inexpensive assay with an automatic analyzer is required. As a result, more effort is needed to translate these findings into more accessible and available clinical tools. In fact, there is currently a significant focus on the spermidine pathway, leading to a strong effort to validate portable methods and sensors that could accurately detect spermidine-related compounds in a robust and easily transferable way; and our results could contribute to accelerate this research.^{33–35}

This study has limitations. The changes in the spermidine pathway concentrations observed are not necessarily specific to CRC and may share aspects of their developmental origins with other physiopathology conditions. Moreover, in this prospective cohort, the number of patients with CRC recurrence was limited.

CONCLUSIONS

Our study provides evidence supporting the association of the spermidine pathway with tumor recurrence following CRC surgery. Confirmation in larger cohorts of patients would allow the translation of this new biomarker into clinical practice.

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DISCUSSANT

Frédéric Ris (Geneva, Switzerland)

I would like to sincerely thank the ESA for the honor of participating in the discussion of this article and the authors for their interesting study. This paper aims to identify biomarkers of recurrence in colorectal cancer, using the spermidine pathway. This pathway is indispensable for cell proliferation and differentiation, and is suggested to accelerate tumor spread. The authors were able to observe, even after a very short follow-up, a difference with a cut-off value comparing the levels of spermidine before and 4 days after surgery. The 16.4% recurrence rate aligns with the current literature and international standards. I first want to congratulate the authors on presenting such impressive data that could lead to a real change of practice in the future, allowing us to assess the risk of recurrence alongside classical risk factors in colorectal cancer.

I have some questions: The first question is, could the spermidine pathway be of interest for a targeted therapy against recurrence? Can we use it to visualize better or as an adjuvant-targeted therapy?

Second, did you notice a difference in the metabolic comparing the local recurrence alone, systemic recurrence, local recurrence, or both?

Finally, do you think this pathway could be applicable in other colorectal cancer metastasis, such as peritoneal carcinomatosis after cytoreductive surgery?

Response From Blanca Montcusí (Barcelona, Spain)

Thank you for your comments. The first question is particularly interesting. This study confirmed an association between the elements from the spermidine pathway and

tumor recurrence, but it is unclear whether this is a cause or an effect. We are not sure if what we observed is a response to cancer aggression. Therefore, to answer this question, we should first study whether there is a direct causal relationship. If this relationship is eventually confirmed, I believe it would be useful for a targeted therapy due to the possibility of controlling spermidine levels. Spermidine is absorbed by the bowel without transformation and can be found in several fruits, vegetables, and fish. There are also spermidine supplements and diets that promote a microbiome capable of producing this element.

Regarding the second question, we did not compare patients with local, systemic, or both recurrences, as we only had two patients with local recurrence. However, I believe it would be interesting to do so if we had more patients with this event.

Finally, with respect to other metastasis such as peritoneal carcinomatosis, I believe this approach could also help in better selecting patients who would benefit from cytoreduction with or without HIPEC (hyperthermic intraperitoneal chemotherapy), for instance.

Thomas Schmidt (Cologne, Germany)

Congratulations on your paper. You showed initially that spermidine was associated with tumor stage. However, I was somehow missing multivariate regression analysis including further clinicopathological parameters including pN category, vascular, and lymph vascular invasion. I am wondering if the data remains of relevance if you include clinical data, which also predicts recurrence. Is the presented data a side effect or the cause of the effect?

Response From Blanca Montcusí (Barcelona, Spain)

I apologize for not being able to respond to this question in detail. Unfortunately, these particular variables were not included in the multivariate analysis.

P. Ronan O’Connell (Dublin, Ireland)

I have two concerns. First, metabolic markers are subject to circadian rhythm and diet, and the question I have for you is how you ensured the baseline was uniform and why you picked day 4 because some patients still are fasting and others will not. Second, why just one sample day 4? You were having these people coming back for regular follow-ups. Why not do sequential follow-up and see if there is a pattern that indicates there is a marker here that we can look at for potential recurrence?

Response From Blanca Montcusí (Barcelona, Spain)

Thank you for your questions. Regarding the first question, we collected all the samples first thing in the morning and following a six-hour of fasting, to control these variations. We selected day 4 because the majority of patients did not undergo postoperative complications, which means they were not fasting and were discharged on this day, being the latest day to easily obtain the sample. Regarding the second question, the aim of this study was to identify patients at high risk of recurrence, but I agree that collecting samples during the follow-up would be of interest to early identify recurrences. This would open up the possibility that the spermidine or an element from its pathway could have a value in disease monitoring, such as CEA (carcinoembryonic antigen). Indeed, we have obtained

samples during the follow-up, which we are now analyzing and hope to achieve results shortly.

Massimo Falconi (Milano, Italy)

I have enjoyed your study. Besides the data that you have presented, the spermidine pathway is a modifiable factor, particularly in patients undergoing neoadjuvant therapy. Did you check whether or not could be changed over the therapy because the data at baseline might be useful in order to assess whether or not to continue neoadjuvant therapy instead or stop?

Response From Blanca Montcusí (Barcelona, Spain)

This is a really interesting question. We excluded patients who had received neoadjuvant therapy, but it would be fascinating to study the behavior of the elements from the spermidine pathway before and after the therapy. In fact, some studies have identified specific metabolites that could be associated with tumor regression grade in locally advanced rectal cancer.

John C. Alverdy (Chicago, USA)

Thank you very much and congratulations on an excellent, lucid, very logical, and clear presentation. I rise to

acknowledge and commend you for bringing up the point that association does not equal causation. I think you dodged the presumptive bullet very cautiously here because, as you know, one cannot modulate something unless it is causative link to what you are observing is established. The question I am asking is related to the fact that we were developing a surgical biobank at the University of Chicago which is a big trend in the United States. The idea is to obtain all the samples that one can procure (exhaled breath, urine, stool, etc.) and put them in a bank and then pair them later to the clinical metadata, as have done here. In your case, the approach is very focused because you are dealing with a pathway that you know plays a role in the phenotype you are describing. What would you do if you had unlimited funds and could collect everything?

Response From Blanca Montcusí (Barcelona, Spain)

I believe the ideal approach would be to collect samples of urine, and specifically stools. Stool samples would provide information about the microbiome, offering a more comprehensive understanding of metabolic phenotyping, which may be influenced by the microbiome. This is something we are currently investigating.